

How do Archaeologists Identify Agriculture?



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Archaeologists have long been interested in the history of agriculture and so have developed a broad suite of methods for studying agricultural practices. These include methods from zooarchaeology and paleoethnobotany (discussed at length later in this module) that allow archaeologists to discern not only what plants and animals were raised, but how people raised those domesticates, and how agricultural economies functioned as a whole.

“Low-Level” Food Production

In an influential article (<https://link.springer.com/article/10.1023/A:1009436110049>) published in 2001, archaeologist and paleoethnobotanist Bruce D. Smith introduces the concept of “low-level” food production. In the article, Smith highlights the idea that agriculture is best seen as one end of a continuum of practices, ranging from occasional cultivation of nondomesticated plants to full-scale, intensive agriculture of domesticates that provides the entire caloric base for a society. In between is a grey area that Smith terms “low-level” food production. This encompasses a variety of practices that involve the cultivation of both morphologically and genetically “wild” species and domesticates, but a significant fraction of calories also comes from gathered plants or hunted animals. Archaeological examples include the early farming communities of North America, which used opportunistic farming of indigenous cultigens (the Eastern Agricultural Complex, or EAC, described by Gremillion beginning on page 59) alongside hunting and gathering of wild plants for thousands of years before the adoption of maize agriculture. A recent ethnographic example is the societies of the American Northwest Coast, who intensively cultivated wild camas plants as a staple starch source and intensively captured and processed wild salmon during spawning season. These rich and reliable food resources allowed large, dense, sedentary populations to form complex social and political systems in the absence of agriculture.

Transhumant Pastoralism

One form of agriculture that was historically common, but is rare today, is transhumant pastoralism, in which people move with the animal herds on which they rely. These animals may be moved on a seasonal basis—for example, into mountains during the summer, when lush plant growth is available, and down to lower elevations during winters, when mountains are icy and impassible. Such patterns are termed seasonal transhumance. An alternative pattern of movement is constant, with no regular, seasonal rotation of areas; this system is often termed nomadic pastoralism. Nomadic pastoralism was historically practiced most often in arid regions in which forage resources and water were discontinuous, separated by less productive deserts. Animals

kept in one location continuously would deplete its resources, so animals were instead moved continually across the landscape, and human populations moved continuously with those animals, never establishing permanent settlements. One famous example of nomadic pastoralism is the Bedouin of Arabia, who moved with their camel and goat herds, living in portable tents.



"A Bedouin Encampment; or, Bedouin Arabs,"
by John Frederick Lewis, painted c. 1841 to
1851.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:John_Frederick_Lewis_-_A_Bedouin_Encampment;_or,_Bedouin_Arabs_-_Google_Art_Project.jpg)

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Agricultural Strategies

Scholars of agriculture define a variety of strategies that farmers employ to manipulate the growing conditions of crops, with the ultimate aims of increasing crop yields and/or reducing the risk of crop failure. How and when these strategies were employed to manage risk and increase yields is discussed in the following section on agricultural intensification and extensification.

Irrigation describes means to deliver supplementary water to crops. This includes everything from hand-watering, as we might in a garden today, to diverting a portion of a river's flow to a field during a critical period of the growth cycle. Typical archaeological evidence for irrigation takes the form of durable remains of irrigation systems, such as the remains of irrigation canals, or evidence from the crops

themselves that the plants received additional water beyond what is possible from local rainfall.

Manuring is a form of fertilization in which animal manure (usually from sheep, goats, or cattle, though human feces have also been used by many cultures, as well as llama dung in the Andes) is mixed with soil in the fields to enrich the soil and improve crop yields. Chemically, this introduces both nitrogen and phosphorus, which are often the two critical elements that, when sufficient amounts are unavailable, most limit plant growth.

Fallow is the period of time in which a field is not farmed, allowing natural vegetation to regrow and reestablish soil nutrients following cultivation. Many fields are fallowed every other year, or for one year in every three or four, especially in cycles of **crop rotation**.

Crop rotation means that different plants are grown in a field in successive planting seasons. For example, modern farmers in the U.S. may plant maize (corn) one year, then soybeans (which fix nitrogen, restoring soil nutrients), and then have a year of fallow before cycling back to maize. Such systems are dependent on knowledge of the characteristics of specific crops and, in periods before people had a detailed understanding of soil chemistry and plant physiology, were developed through trial and error over centuries, which resulted in sophisticated systems of traditional farming.

Polycropping (also termed multicropping) is planting multiple crops together, either at once or in sequence, in a single field. For example, maize, beans, and squash (known together as the “Three Sisters”) were often planted together in traditional agricultural systems of eastern North America. Each plant supported the growth of the others: maize grew tall, providing a scaffold for bean vines to grow upward to the sun; beans supplied nitrogen to the soil; and squash grew broad leaves close to the ground that shaded the soil, maintaining soil moisture and limiting weed growth.